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RESEARCH

Quantifying the impact of multiple stressors on microbial communities in Dianshan Lake sediments using Random Forest analysis

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Abstract Alluvial plain lakes, characterized by low hydrodynamic activity and fine sediments, host microorganisms whose diversity and community structure are strongly shaped by the presence of nutrients and pollutants. However, quantifying the complex impacts of these multiple stresses is challenging. This study focused on Dianshan Lake, a human-impacted plain lake in Shanghai, China, to quantify the contributions of sediment properties, pollution levels, and nutrients on microorganisms, using classical statistical methods and Random Forest (RF) analysis. The RF model achieve a good fit ($R^2 = 0.80–0.87$). Feature contribution analysis indicated that contaminants were the primary factors shaping microbial diversity in sediments, accounting for approximately 40% of the total contribution. However, sediment redox conditions emerged as the most influential single factor. In lightly to moderately polluted freshwater lakes, benthic microorganisms displayed common dominance, but the contributions of influencing factors varied. Proteobacteria and Chloroflexi, the dominant phyla, were significantly impacted by pollutants, with contributions exceeding 50%. PAHs (Polycyclic Aromatic Hydrocarbons) primarily suppressed genera within Proteobacteria, while *Anaerolineaceae* in Chloroflexi exhibited strong tolerance to Cd. The dominant species in Dianshan Lake sediments were also strongly influenced by NO_3^--N and NH_4^+-N, far exceeding the impact of various forms of phosphorus. This also highlights the issue of nitrate-driven eutrophication in the region. This study demonstrates that RF analysis effectively identifies key controlling factors in lightly to moderately polluted sedimentary environments, providing valuable insights into the ecological processes and a scientific foundation for ecological risk management in similar aquatic environments.

Keywords Benthic microorganisms · Environmental stressors · Potential toxic elements · Nutrients

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Introduction

Lake sediments and their associated microorganisms are integral components of aquatic ecosystems, playing pivotal roles in the biogeochemical cycling of essential elements (Rathour et al., 2020). The physicochemical properties of sediments—such as pH, particle size distribution, redox conditions, and habitat heterogeneity—are important drivers of microbial diversity and community composition (Manirakiza et al., 2022). In freshwater lakes, sediments often act as sinks for a wide range of contaminants, including toxic metals, polychlorinated biphenyls (PCBs), and excess nutrients. Microbial communities inhabiting these sediments are consequently exposed to multiple anthropogenic stressors (Allan, 2004). When combined with natural environmental fluctuations, these stressors can substantially reshape microbial community structure and function, with cascading effects on ecosystem health and resilience (Chen et al., 2024). Consequently, growing research attention has been directed toward elucidating the key environmental factors that most significantly influence microbial diversity within sediment ecosystems.

The physicochemical properties of sediments, such as dissolved oxygen and pH, play critical roles in shaping the types and spatial distribution of microorganisms. Aerobic microorganisms typically thrive under redox potentials (Eh) ranging from +300 to +400 mV, whereas anaerobic microorganisms favour reducing conditions, with Eh < +100 mV (Hunting & Kampfraath, 2013). Additionally, pH also exerts a direct impact on the relative abundance of certain bacterial taxa, thereby affecting the overall microbial community structure (Lauber et al., 2009). In general, stable sediments pH values between 7.4 and 7.8 can maintain intracellular pH (Padan et al., 2005), while higher pH condition (>7.8) may compromise bacterial survival (Tripathi et al., 2018). Xiong et al. (2012) further observed that in sediment from 15 lakes in the Tibetan Plateau, pH was inversely correlated with microbial α-diversity and significantly influenced the abundance of δ-Proteobacteria. Moreover, sediment particle size distribution and porosity can indirectly shape microbial diversity and community composition by regulating nutrient adsorption, gas exchange, and other physicochemical processes (Chen et al., 2022; Juvigny-Khenafou et al., 2021; Lu et al., 2022).

In addition to physicochemical factors, the availability of nutrients—such as organic carbon, nitrogen, phosphorus, and sulfate—also plays a pivotal role in shaping microbial growth and community composition within sediments (Zhang et al., 2015a; Fan et al., 2019; Prasath et al., 2021; Liao et al., 2021). Under eutrophic conditions, excessive nitrogen and phosphorous inputs can lead to an increase in microbial biomass; however, this is often accompanied by a reduction in microbial diversity and a shift towards more homogeneous bacterial and fungal communities (Yu et al., 2021; Geng et al., 2022; Sheng et al., 2011). Furthermore, excessive organic carbon can deplete dissolved oxygen, thereby inhibiting the growth of aerobic organisms like nitrifying bacteria. Sulfur-oxidizing bacteria can also alter microbial community composition by modifying pH and redox conditions, thereby influencing the overall ecological balance (Hu et al., 2018; Jeppesen et al., 2005; Zhang et al., 2024e).

Anthropogenic pollutants can substantially inhibit microbial growth and activity (Yan et al., 2024). Numerous studies have reported negative correlations between pollutant concentrations and microbial biomass or diversity (Gao et al., 2024; Jiang et al., 2016; Wallberg & Andersson, 1999). However, the relationship between pollutants and microbial diversity is not always linear (Pan et al., 2022). For instance, low concentrations of Sb and As (<80 mg/kg) can even stimulate bacterial growth (Wang et al., 2023a). Some members of Proteobacteria, such as *Pseudomonas syringae*, carry copper resistance genes that confer tolerance to metal stress (Mellano & Cooksey, 1988). In addition, several bacterial genera—such as *Alicyclobacillus, Brevibacterium,* and *Lactobacillus*—have demonstrated the capacity to degrade toxic organic pollutants, including polycyclic aromatic hydrocarbons (PAHs) (Li et al., 2019). As a result, pollution-tolerant microorganisms often become dominant in heavily contaminated sediments, while more sensitive taxa are suppressed (Rahman, 2020; Ramsay et al., 2000; Wang et al., 2022b).

Given the complexity of factors influencing sedimentary microbial communities, this study employs a typical multifunctional lake as a case study and applies the Random Forest (RF) algorithm to quantify the relative effects of multiple environmental drivers on microbial diversity and community composition under varying pollution pressures. We hypothesize

that in lightly to moderately polluted sediments, the controlling factors governing microbial diversity and dominant taxa differ substantially, with distinct contributions from sediment properties, nutrient levels, and contaminant concentrations.

The case study site, Dianshan Lake—the largest freshwater lake in Shanghai, China—serves multiple ecological and socioeconomic functions, including drinking water supply, navigation, tourism, and fisheries. Environmental conditions in the lake directly affect the safety of drinking water for surrounding communities. Although extensive research has addressed eutrophication, organic contamination, and heavy metal pollution in Dianshan Lake, limited attention has been given to the sedimentary microbial communities and the factors shaping their distribution.

Machine learning techniques have recently been applied to investigate the ecological impacts of environmental variables (Grekousis et al., 2022; Heidari et al., 2023; Sa'adi et al., 2024; Zhang et al., 2024d), with the RF algorithm showing particular robustness and interpretability in such analyses (Zhang et al., 2024c; Ramos Collin et al., 2024; Cardiology et al. 2023). In this study, high-throughput sequencing was used to characterize the microbial diversity and community structure of surface sediments collected from different regions of Dianshan Lake. By integrating RF modeling with sediment physicochemical data, nutrient concentrations, and contaminant levels, we aim to identify and quantify the key environmental factors shaping sediment microbial communities. The findings provide theoretical support for water quality management and environmental protection in Dianshan Lake and offer broader implications for similar freshwater systems.

Materials and methods

Site description and sediment sampling

Dianshan Lake is located in the suburban area of Shanghai, adjacent to Jiangsu Province (30°40′–31°53′N, 120°52′~122°12′E). It is the largest natural freshwater lake in Shanghai, covering approximately 63 km^2, with an average depth of 2.1 m (Meng et al., 2020). Situated on an alluvial plain, the lake is part of the extensive river network downstream of the Yangtze River, connecting to 59 rivers. The inflowing rivers are predominantly located on the west and north sides of the lake, while the outflows are mainly on the east side, with Lanlu Port accounting for 71% of the total outflow (Zhou, 2022). The maximum vertical current speed ranges between 0.02 and 0.05 m/s (Ming-Wei et al., 2011), thus minimizing the impact of flow rate on the spatial distribution of sediment microbes.

Following field investigations and surveys, a total of 16 sampling points were selected in November 2022 (Fig. 1). Surface sediment samples were collected using a sediment core sampler. After allowing the samples to settle, non-decomposed litter was removed, and the overlying water was discarded. The sediment samples were then sealed in sterilized polyethylene bags, labeled with corresponding sampling point numbers, placed in a thermal insulation box with ice packs, and immediately transported to the laboratory for further analysis.

Sediment property analysis

All samples were air-dried and passed through a 2.0 mm nylon sieve for storage and subsequent physicochemical analysis. Sediment pH was determined using a pH meter with a combined glass electrode (Mettler Toledo, PE 20) in a suspension of sediment and deionized water at a ratio of 1 g: 2.5 mL (Yan et al., 2018). The particle size distribution was analysed using a laser diffraction particle analyser (Baite, BT-9300Z), with particle classification based on Wentworth's Sedimentary Scale (Wentworth, 1922). The dried samples were carefully ground to particles smaller than 0.15 mm to measure the total organic carbon (TOC) content, using a total organic carbon analyzer (TOCVCPN, Shimadzu, Japan) (Muñoz et al., 2015).

Analysis of contaminants in sediments

The concentrations of nine potential toxic element (PTE) elements (Cd, Co, Cr, Cu, Mo, Ni, Pb, Sb, and Zn) were measured in 0.1 g of finely ground samples (<75 μm). The samples were digested using a mixture of 1 mL H_2O_2 (30%), 1 mL HF (40%), and 3 mL HNO_3 (65%) in sealed Teflon vessels within a microwave digestion system (PreeKem, TOPEX). For samples with high total organic carbon (TOC) content, an

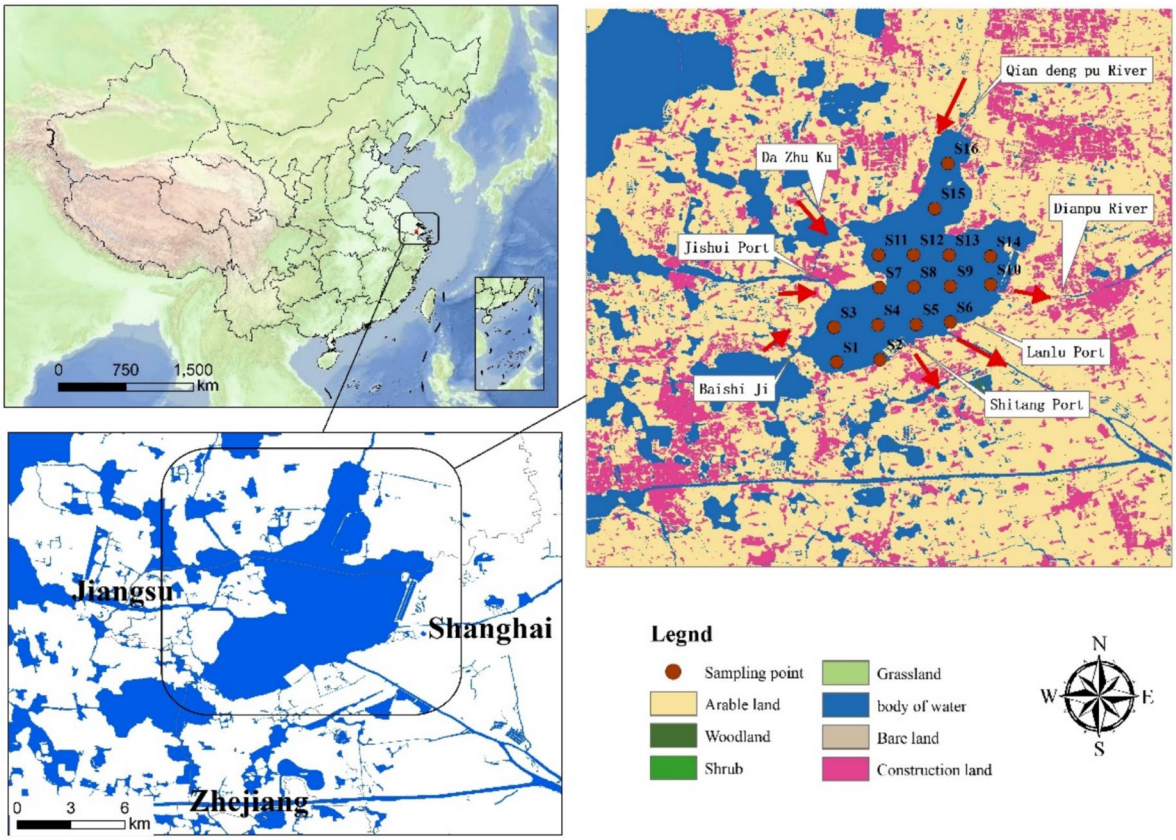

Fig. 1 Geographic map of the Dianshan Lake study area, showing locations of surface sediment sampling (top 10 cm). Dianshan Lake is displayed in the context of three neighbouring administrative regions: Jiangsu, Shanghai, and Zhejiang. Red arrows indicate the main inflowing and outflowing rivers

additional 1 mL of $HClO_4$ was included. The digests were then heated on a hot plate (~150 °C) to drive away the $HClO_4$ before being diluted to 50 mL in a volumetric flask with Milli-Q water and 5% HNO_3 (Mao et al., 2022). Element concentrations were analysed using inductively coupled plasma mass spectrometry (ICP-MS, PerkinElmer, NexIon 300X).

Quality assurance and quality control (QA/QC) procedures for PTE concentration analysis included the use of reagent blanks and standard reference materials (SRM 1646a, National Institute of Standards and Technology). To prevent contamination, all acids used in the experiment were of Guaranteed Reagent grade. Containers were soaked overnight in 10% (v/v) diluted HNO_3 and rinsed thoroughly with diluted HNO_3 and Milli-Q water before use. Replicates were analysed every eight samples, and the relative standard deviation (RSD) among replicates was consistently below 10.8%. The recovery rates for selected PTEs ranged between 80 and 120%.

Analysis of nitrogen and phosphorous in sediments

Total nitrogen was (TN) determined on dried and homogenized sediments using an automatic Kjeldahl apparatus according to the method described in China's *Soil testing Part 24: Determination of total nitrogen in soil, Automatic nitrogen determinator method* (*NY/T 1121.24–2012.24*) (Ministry of Agriculture 2012). Exchangeable NH_4^+-N and NO_3^--N were determined according to the extraction method described in Fan et al. (2018). The concentrations were measured by a spectrophotometer. Total phosphorus (TP) was determined by solubilizing all phosphorus in the sediment using NaOH at high temperatures (400 °C and 640 °C), with concentrations

measured spectrophotometrically at 700 nm, the method described in China's *Soil—Determination of Total Phosphorus by alkali fusion–Mo–Sb Anti spectrophotometric method (HJ 632–2011)* (Ministry of Ecology 2011). Phosphorus fractionation, including organic phosphorus (OP), apatite (AP), and non-apatite inorganic phosphorus (NAIP), was measured according to the Standards, Measurements and Testing programme (SMT) protocol (Ruban et al., 1999). The recovery of P fractionation was in the range of 95~103%.

DNA extraction, PCR amplification, and sequencing

Microbial DNA was extracted from sediment samples using the FastDNA® Spin Kit for Soil (Omega Bio-tek, Norcross, GA, U.S.), following the manufacturer's protocol with careful attention to sterility throughout the process. The purified DNA was dissolved in 200 μL of TE buffer (pH 8.0) and immediately placed in a thermally insulated container with ice packs for transport to Shanghai Meiji Biomedical Technology Co., Ltd. (Meiji Biotech) for 16S rDNA gene sequencing. PCR amplification (T100 Thermal Cycler, BIO-RAD, U.S.) targeted the V3-V4 hypervariable region of the 16S rDNA gene using universal bacterial primers 338 F (5′-ACTCCTACGGGAGGC AGCAG-3′) and 806R (5′-GGACTACHVGGGTWT CTAAY-3′). High-throughput sequencing was then conducted on the Illumina MiSeq PE300 platform at Meiji Biotech.

The resulting sequencing data were aligned with a reference database to generate an Operational Taxonomic Unit (OTU) classification table, providing a detailed profile of the microbial community structure. Using UPARSE v11.0.667, sequences were clustered into OTUs at 97% similarity, and chimeras were removed (Edgar et al. 2013; Stackebrandt & Goebel, 1994). All samples were rarefied to 20,000 sequences to minimize the effects of sequencing depth on diversity analyses, retaining 99.09% Good's coverage. OTUs were taxonomically classified against the Silva database (v138.2) using the RDP classifier (http://rdp.cme.msu.edu/, version 2.11) (Wang et al. 2007) with a 70% confidence threshold, and community composition was analyzed across multiple taxonomic levels. To minimize the influence of uneven sequencing depth among samples, the OTU table was rarefied to the minimum sample size before α- and β-diversity analyses.

Statistical analysis and GIS mapping

This study selected the Sobs, Heip, Shannon, and Simpson indices to analyse the richness, evenness, and diversity of microbial communities at the OTU level in sediment, while the Coverage index was used to assess the accuracy of sequencing results. Beta diversity analysis focused on comparing the diversity between different habitats or microbial communities to explore the differences in community structure composition between samples (Li et al., 2022). Microbial community similarities among samples were evaluated using Non-metric Multidimensional Scaling (NMDS) based on Bray–Curtis dissimilarity (Vegan v2.5–3) (Huang et al., 2023a, 2023b). To address potential multicollinearity among environmental factors, we calculated the variance inflation factor (VIF) for each variable. We followed an iterative process by sequentially removing the variable with the highest VIF until all remaining variables had a VIF value below 10 (Jiao et al., 2025). This procedure resulted in the selection of key environmental variables for subsequent constrained ordination analyses. Redundancy Analysis (RDA) was conducted to examine the relationships between community composition and environmental variables. The potential relationship between environmental factors was investigated using Spearman correlation analysis. Co-occurrence network of Microbial Communities was analysed using the MAJORBIO Bioinformatics Cloud Platform (https://cloud.majorbio.com/), the *P*-value is lower than 0.05. Constructing a co-occurrence network using the Fruchterman Reingold layout algorithm with Gephi 0.9.2 software. The topological characteristics of the co-occurrence networks were analysed, including the positive and negative correlation numbers, as well as the numbers of edges and nodes. Area digitization and mapping were performed using ArcGIS 10.6.

The Random Forest (RF) model, proposed by Breiman in 2001, is a machine learning algorithm based on decision trees (Breiman, 2001). RF uses bootstrap resampling to extract and generate subsets of the training sample from the original training dataset. It then builds multiple decision trees based on these training subsets and forms a random

forest, with classification or regression model outcomes determined by the voting scores of the decision trees (Breiman, 2001). In this study, an RF model was developed using data that included the physicochemical properties, nutrients, and pollution factors of Dianshan Lake sediments, as well as microbial diversity (Shannon index), and the relative abundances of the five dominant phyla. Environmental factors from 16 sampling points were treated as independent variables, while Shannon diversity and species abundance indices were treated as dependent variables. The RF simulation was performed using the custom software PyCharm, and the importance of pollution factors on microbial diversity and abundance was assessed. The variables were ranked according to their importance in the model, and the accuracy of the model was evaluated using the correlation coefficient (R^2) and the root mean square error (RMSE) values. A flowchart of the machine learning analysis process is provided in Figure S1 and text S1 of the supplementary material. The performance of the Random Forest model was internally validated using the Out-of-Bag error. Additionally, cross-validation was employed to further assess the model's prediction accuracy and robustness. Detailed validation results are provided in Table S5. To enhance interpretability, the SHapley Additive exPlanations (SHAP) was also used to systematically quantify the influence of environmental parameters on the microbial diversity and community within the study area. The standardization of environmental variables has a negligible impact on the Random Forest model and SHAP results. Relevant comparisons are provided in the figures and tables included in the supplementary material (Figure S2 and Table S1).

In addition, to assess the extent of anthropogenic pollution and ecological risk of PTEs in surface sediments, this study also applied the geoaccumulation index (I_{geo}, Müller, 1969) and the Potential Ecological Risk Index (PERI, Kong et al., 2011; Yunker et al., 2002). The detailed calculation methods are provided in Supplementary Text 2. Furthermore, we employed the characteristic molecular ratios approach for the diagnosis of PAH sources in the surface sediments of Dianshan Lake.

Results

Environmental factors in Dianshan Lake sediment

Sediment characterization

The physicochemical characteristics of sediment samples from Dianshan Lake are summarized in Table 1.(a). The sediment composition predominantly comprised silt and clay, with a relatively low sand fraction. The coefficients of variation (CV) for grain size distributions indicate that silt, which constitutes the majority, exhibited minimal variation, suggesting an even grain size distribution across the study area. The average pH was 7.73, with a range of 7.33 to 8.07, reflecting a weakly alkaline environment. The CV of pH was 2.9%, indicating minimal variation. Lower pH values were observed near inflow channels, while higher values were concentrated in the northeastern part of the lake and around discharge areas such as Lanlu Port and the Dupu River. The average redox potential (Eh) was +258 mV, varying between +219 and +285 mV, indicating a weakly oxidizing environment for surface sediments. As a typical shallow lake with lower clay content, the organic matter in these silty sediments was less than that found in clay-rich sediments, making them more susceptible to influences from overlying water, which may hinder the metabolic activities of anaerobic bacteria. This shallowness contributes to elevated sediment Eh values, with a CV of 6.7%. As illustrated in Figure S3, higher Eh values were concentrated near inflow channels, while lower values are predominantly found in the northeastern region of the lake.

Nutrient levels

The concentrations of nitrogen and phosphorus species in the surface sediments of Dianshan Lake are detailed in Table 1.(a). The distribution of these elements was notably uneven, with CV exceeding 20%. The concentrations of NH_4^+-N and NO_3^--N ranged from 35.47 to 95.37 mg/kg and 3.58 to 13.27 mg/kg, respectively, with higher concentrations located near inflow channels and a general decrease observed downstream. Overall, IP concentrations surpassed those of OP, with NAIP constituting approximately 69% of total inorganic phosphorus content. Elevated concentrations of phosphorus forms were primarily

Table 1 Summary of the a) sediment properties and b) concentration of contaminants in Dianshan Lake sediments, along with data from other lakes

Location	pH	Eh (mV)	Sand (%)	Silt(%)	Clay (%)	TOC (%)	NO_3^--N (mg/kg)	NH_4^+-N (mg/kg)	TN (mg/kg)	OP (mg/kg)	NAIP (mg/kg)	AIP (mg/kg)	TP (mg/kg)	Reference
DianshanLake China	7.73 (2.9%)*	258 (6.7%)	3.3 (108%)	81.2% (3.5%)	15.6 (19.8%)	1.3 (20.3%)	7.18 (34.4%)	68.65 (26.1%)	483.29 (22.9%)	278.23 (26.1%)	209.45 (26.4%)	94.57 (27.8%)	669.89 (27.3%)	This study
Taihu Lake (China)	7.1	145.93				2.35	6.45	82.53	1315.50	172.557			878.169	[63, 64]
Chaohu Lake (China)	6.3							29.52	532	70.59		12.28	358	[65]
Lake Kirkkojärvi (Finland)	7.5					11	434.6	58.6	1383.3				73.3	[66]
Lake Bützow (Germany)			15	35.7	17.7	11.8		1.886	13000	170	169	78	1520	[67]
Location	Cd (mg/kg)	Co (mg/kg)	Cr (mg/kg)	Cu (mg/kg)	Mo (mg/kg)	Ni (mg/kg)	Pb (mg/kg)	Sb (mg/kg)	Zn (mg/kg)	∑16PAHs (ng/g)	Reference			
Dianshan Lake (China)	0.18 (34.2%)	13.72 (14.5%)	62.93 (11.9%)	49.54 (36.4%)	0.59 (23.7%)	46.59 (18.3%)	26.51 (15.8%)	2.32 (21.0%)	119.28 (24.7%)	36.32 (6.06%)	This study			
Taihu Lake (China)	0.52		80.13	33.81		39.26	32.27		102.99	102.0	[63]			
Chaohu Lake (China)	0.77		67.1	33.0		28.7	54.1	1.45	306.0	334.67	[68, 69]			
Lake Erie (America)	1.5	14.0	51.8	47.1	1.7	53.4	43.4		156.3	420.0	[70,71]			
Lake Srebarna (Bulgaria)	0.6		117	52.6		47.7	183		130	2556	[72]			
Lake Nicaragua(Nicaragua)	0.15		39	130			7.0		89	640	[73]			

*numbers in brackets are the coefficient of variance.

found in proximity to inflow channels, notably Qian Deng Pu and Da Zhu Ku. Based on Table 1.(a), it can be concluded that while Dianshan Lake shows signs of eutrophication, its severity is relatively lower compared to other alluvial lakes worldwide.

Level and distribution of contaminants in surface sediment

Table 1.(b) presents the concentrations of 9 PTEs (Cd, Pb, Sb, Cr, Ni, Cu, Zn, Co, and Mo) in the surface sediments of Dianshan Lake. Only Cr and Co exhibited concentrations below the background values established for the Taihu lake plain (Yin et al., 2011). Results from the geoaccumulation index (Table S2) indicate that, on average, Cd, Cu, Sb, and Zn reach a weakly polluted level, indicating anthropogenic input on these elements. On the other hand, the PERI (Figure S4) shows that Cd and Sb pose a relatively high ecological risk at some sampling sites and contribute the most to the total risk index, with most sites falling within the moderate risk category. The spatial distribution of these 9 PTEs demonstrates a consistent pattern, with higher concentrations observed near inflow channels and a gradual decline downstream. Notably, elevated levels of Cu and Mo were concentrated in the northern region near the inflow river of Qian Deng Pu, while Cd, Pb, and Co were enriched in inflow areas on the western side of the lake. Furthermore, the distributions of Sb, Cr, Ni, and Zn displayed similarities, with high concentrations found in both the northern and western inflow regions, including sites such as Baishi Ji, Jishui Port, and Da Zhu Ku. As shown in Fig. 1, the northern region is primarily an industrially dense area, while the western region is a mixed area of residential and agricultural zones. A comparison of the PTE contents in the sediments of other lakes, as shown in Table 1.(b), reveals that, except for slightly elevated levels of Cu, Ni, and Sb, the concentrations of all other PTEs are lower than those in the other lakes.

In comparison to other aquatic environments, the total concentration of 16 polycyclic aromatic hydrocarbons (Σ16PAHs) in the surface sediments of Dianshan Lake ranged from 7.54 to 128.94 ng/g, with an average concentration of 36.32 ng/g (Table 1.(b)). All 16 PAHs were detected, with detection rates and concentration ranges provided in Table S3. Notably, PAHs such as Phe, Flt, Pyr, BaA, Chr, BbF, BkF, BaP, and InP were consistently present, each exhibiting a 100% detection rate. Among the 16 PAHs, Flt, Pyr, and BaP exhibited the highest average concentrations at 5.47, 5.30, and 4.93 ng/g, respectively. The highest concentrations of PAHs were primarily located near the inflow channels in the northern region, with additional accumulation observed in the southwestern part of the lake (Fig. 2). As it has been described above, the northern region is mainly an industrial area. Based on the data comparison in Table 1.(b), the PAHs content in Dianshan Lake is significantly lower than that in the sediments of other lakes, indicating that the PAHs pollution in Dianshan Lake is not severe.

Microbial community in Dianshan Lake sediment

α and β biodiversity

The α-diversity analysis across the study area revealed that the Coverage indices for all sampling sites > 0.96, approaching 1.0, indicating sufficient sequencing depth and reliable results. The Sobs, Heip, Shannon, and Simpson indices varied from 2334 to 2829, 0.169 to 0.245, 6.005 to 6.541, and 0.00474 to 0.01079, respectively, with mean values of 2493, 0.209, 6.251, and 0.00666 (Supplementary Material, Figure S5). The Sobs index reflects species richness, while the Heip index indicates community evenness. The Shannon and Simpson indices combine both richness and evenness to depict overall community diversity. Among them, the Shannon index is more responsive to variations in species richness, whereas the Simpson index places greater emphasis on dominant species. The spatial distribution of the Sobs, Heip, and Shannon indices exhibited consistent patterns, with higher values predominantly located near the outflow river channels and the western inflow areas. These results suggest that these regions are characterized by elevated microbial richness, evenness, and diversity. Conversely, lower values were observed in the central and eastern parts of the lake, as well as around the Qiandengpu River inflow in the northern region, indicating lower diversity in these areas.

Non-metric multidimensional scaling (NMDS) of the spatial relationships between sampling points further delineated the study area into four distinct zones (Fig. 3). The NMDS results yielded a stress value of 0.107, suggesting that the two-dimensional ordination

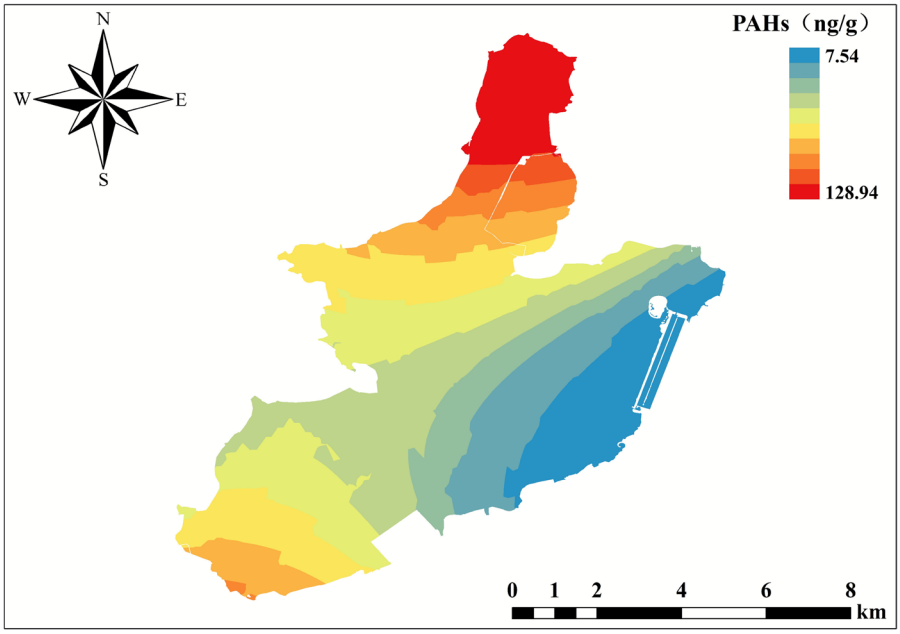

Fig. 2 Distribution of 16ΣPAHs (ng/g) in surface sediment from Dianshan Lake

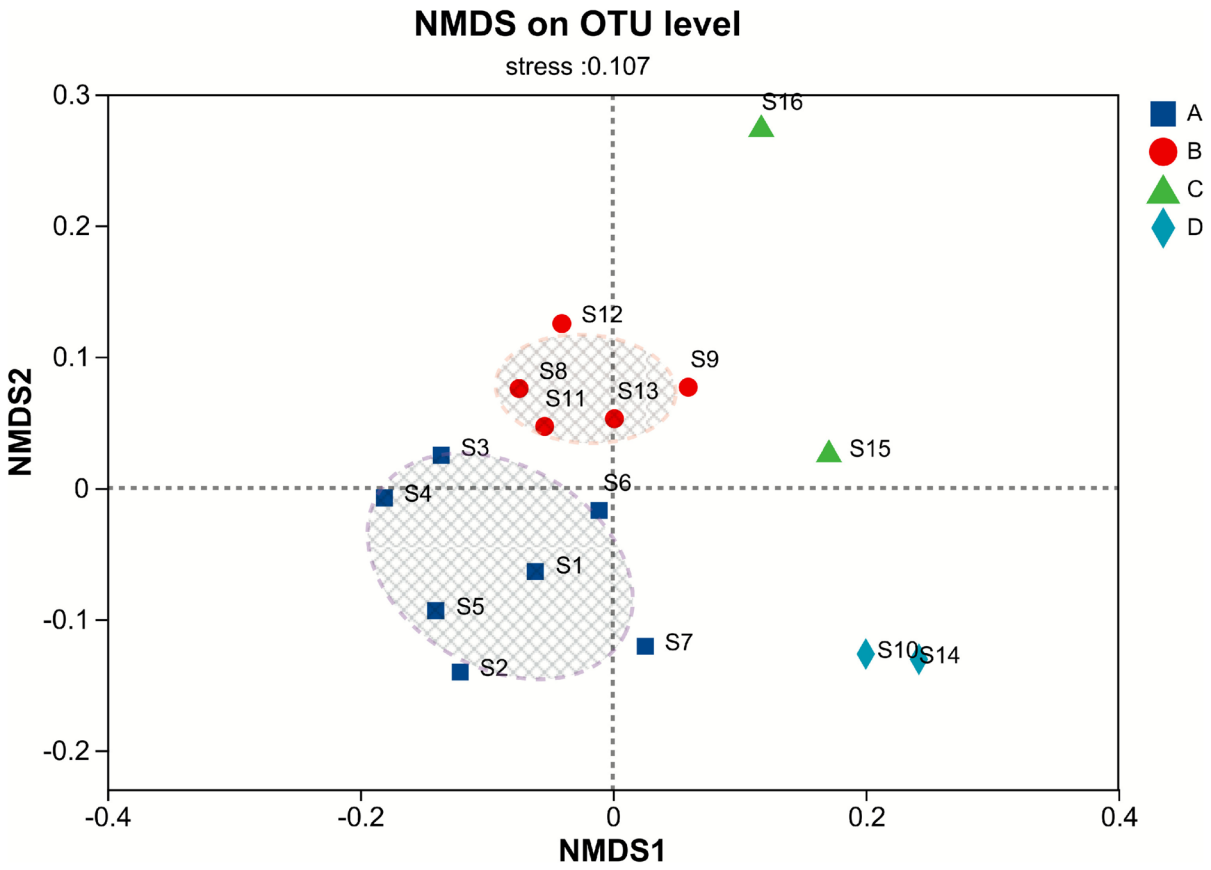

Fig. 3 Non-metric multidimensional scaling (NMDS) revealed the separation and aggregation of microbial communities from 16 sampling sites across the four regions of Dianshan Lake

provides an acceptable representation of the community structure differences. The results also show that these microbials were grouped according to their sampling location. Zone A, comprising sites S1–S7, was primarily located in the southern and southwestern parts of the lake, a region characterized by two navigational channels and surrounded by villages and agricultural land. Zone B encompassed the central lake region, including sites S8, S9, and S11–S13. Zone C, represented by sites S15 and S16, was situated in the northern part of the lake, receiving inflows primarily from Qiandengpu River. Zone D, which included sites S10 and S14, was located in the western part of the lake, near construction and the outflow river channels. Samples from Zones A, B, and D exhibited relatively close clustering, whereas S15 and S16 showed notable dissimilarity.

Micro-organism community structure on OTU level

A total of 5717 OTUs were identified from 16 surface sediment samples. Based on database comparison, these OTUs were classified into 60 phyla, 174 classes, 367 orders, 534 families, 828 genera, and 810 species within the bacterial domain. Figure 4a presents a Venn diagram of species from the 16 sampling sites at the OTU level, allowing for a comparative analysis of the shared and unique OTUs across different sampling sites. The diagram shows that while there are variations in the number of unique OTUs across the sites, the differences are generally small. Sampling site S3 had the fewest unique OTUs, with only 22, whereas site S16 had the most unique OTUs, indicating a higher presence of site-specific microbial species.

Figure 4b illustrates the microbial community composition at the phylum level for species with a relative abundance ≥1% in the sediment of Dianshan Lake. While the types of microbial species were consistent across the sampling sites, their relative abundances varied significantly. Specifically, the surface sediments at each site were primarily dominated by Chloroflexi (14.91%–35.21%), Proteobacteria (17.13%–31.07%), Acidobacteriota (4.78%–21.64%), Actinobacteriota (6.41%–11.87%), Desulfobacterota (3.28%–7.87%), Nitrospirota (2.90%–8.58%), Firmicutes (2.15%–7.72%), and Bacteroidota (2.13%–4.90%).

At the genus level, a large proportion of the microbial genera had a relative abundance of less than 1%, with these low-abundance genera accounting for more than 35% of the total abundance (Figure S6). This indicates a high genus-level richness in the sediments of Dianshan Lake, consistent with the results of the α-diversity analysis. The two most dominant genera in the sediments were from the

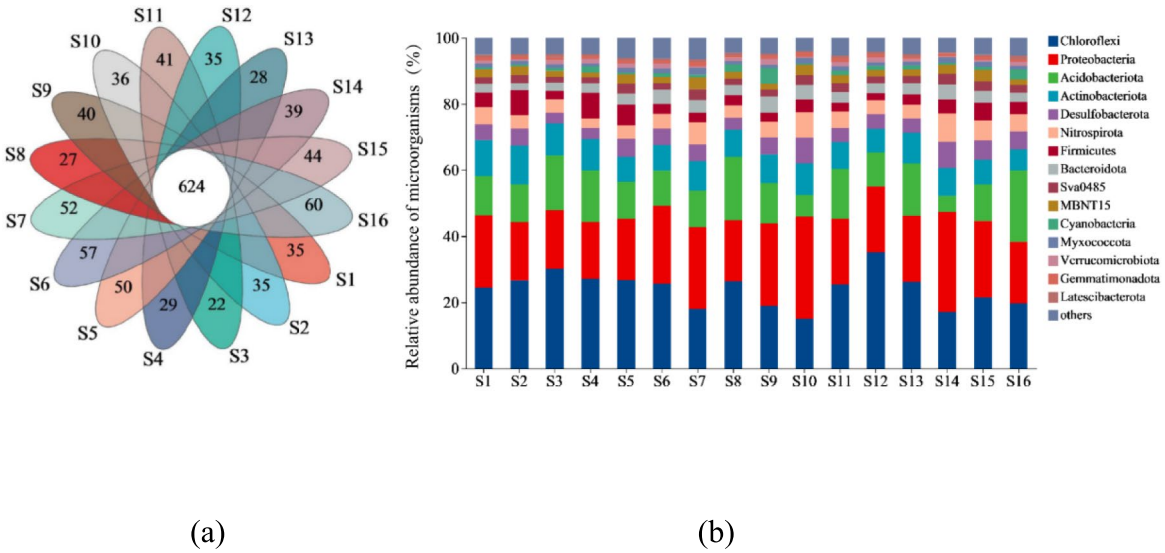

Fig. 4 **a** Venn diagram of sediment species at each site based on OTU levels. **b** Relative abundance of surface sediment microorganisms at the phylum level across sampling sites in Dianshan Lake

family *Anaerolineaceae*, with relative abundances ranging from 3.60% to 24.77%, and from the class *Thermodesulfovibrionia*, with relative abundances ranging from 2.20% to 7.01%. Although the microbial genera were similar across the sampling sites, the relative abundance of each genus varied across the sites.

Co-occurrence network of microbial communities

The study analysed the co-occurrence network of the top 50 dominant microorganisms at the genus level, along with their key topological properties, to elucidate the complex interactions among microorganisms and reveal the relationships between key species in the environment (Fig. 5 and Table S4). In the microbial network of Dianshan Lake sediment, a total of 369 edges were identified, with 199 positive connections, accounting for 53.9%. The network was dominated by positive correlations, indicating potential cooperative or functionally synergistic relationships among microbial taxa, possibly through shared metabolic pathways or similar environmental adaptations (Faust & Raes, 2012). As the network was constructed from compositional data without applying methods such as SparCC or SPIEC-EASI, the inferred associations should be interpreted with caution. Negative correlations may suggest competitive interactions for limited resources such as space and nutrients (Fuhrman & Steele, 2008). The relatively high average degree (14.76) and clustering coefficient (0.612) indicate a densely connected and complex network structure, in which nodes with higher connectivity may play more central ecological roles (Rong et al., 2021).

As shown in the figure, *Anaerolineaceae* (Chloroflexi) exhibits the highest number of connections with other microorganisms, followed by *Vicinamibacterales* (Acidobacteriota), *SC-I-84* (Proteobacteria), *MB-A2-108* (Actinobacteriota), and *Sva0081* (Desulfobacterota), all of which also demonstrate significant interactions with other microorganisms. This highlights their role as key species within the network. Notably, *Thermodesulfovibrionia* (Nitrospirota), although exhibiting many connections with other species, is not among the most abundant species in the network.

Relationship between environmental factors and microbial communities

Correlation between environmental factors and microbial communities

To better understand the relationships between microbial communities in surface sediments of Dianshan Lake and individual environmental factors, Spearman correlation analyses were conducted. These analyses assessed the relative abundances of the top 10 dominant phyla (Fig. 6a) and the top 20 dominant genera (Fig. 6b) against sediment physicochemical properties, concentrations of contaminants, and nutrient contents. At the phylum level, negative correlations with environmental factors were more frequently observed. Total nitrogen (TN) and NO_3^--N emerged as key factors, influencing the largest number of dominant phyla. Specifically, Bacteroidota, Desulfobacterota, and Sva0485 were significantly negatively correlated with TN ($p < 0.05$). Among the phyla, Bacteroidota and Acidobacteriota were the most affected by individual environmental factors, although their response patterns were entirely opposite. In terms of pollutant influence, PTEs affected half of the dominant phyla. Notably, Cd was significantly positively correlated with Chloroflexi ($p < 0.05$), while Sb, Cu, Zn, and Co were significantly negatively correlated with other dominant phyla. Additionally, PAHs exhibited a significant positive correlation with Acidobacteriota ($p < 0.05$) and a significant negative correlation with Bacteroidota ($p < 0.05$).

At the genus level, all environmental factors, except for TOC, Pb, and Mo, were significantly correlated with the relative abundances of microbial taxa. Overall, nitrogen species had the most pronounced effect on sediment microbial communities, influencing 8 dominant genera, most of which displayed significant positive correlations. *Vicinamibacterales* and *Luteitalea* were strongly correlated with NH_4^{+-}N, with R^2 of 0.78 and 0.75, respectively, while *MB-A2-108* was strongly correlated with NO_3^--N ($R^2 = 0.77$). Particle size distribution was significantly associated

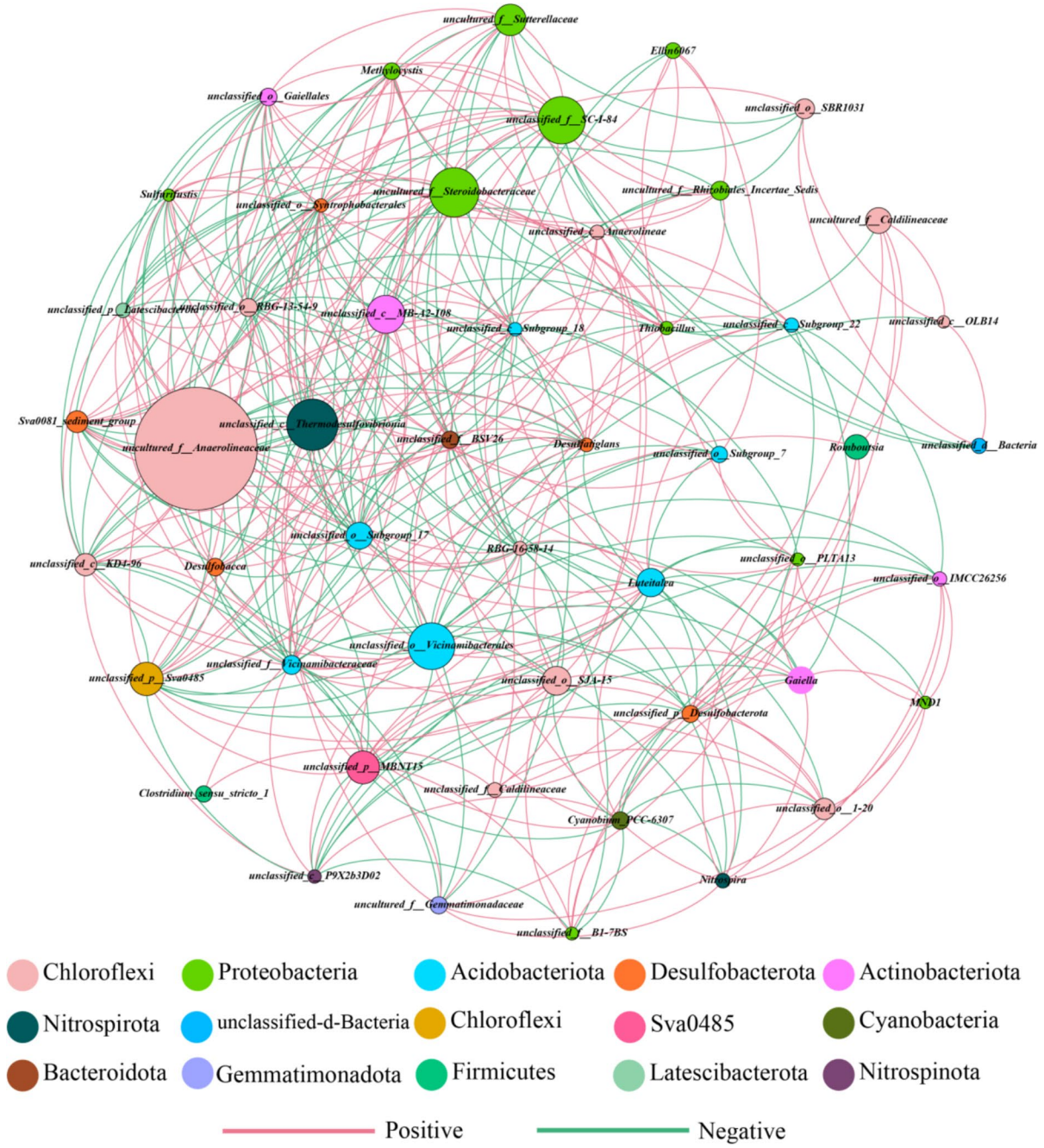

Fig. 5 Co-occurrence networks of microbial community at the top 50 genera in sediments. The size of each node is proportional to the connectivity degree. The colours of the nodes represent different phyla. Red and green edges represent positive and negative correlations, respectively

with five genera. Regarding pollutant effects, PAHs and multiple PTEs demonstrated significant correlations with microbial genera, but these effects varied between taxa. Phosphorus primarily impacted *unclassified_o__1-20*, a genus within *Steroidobacteraceae*, and *SJA-15*, with all three

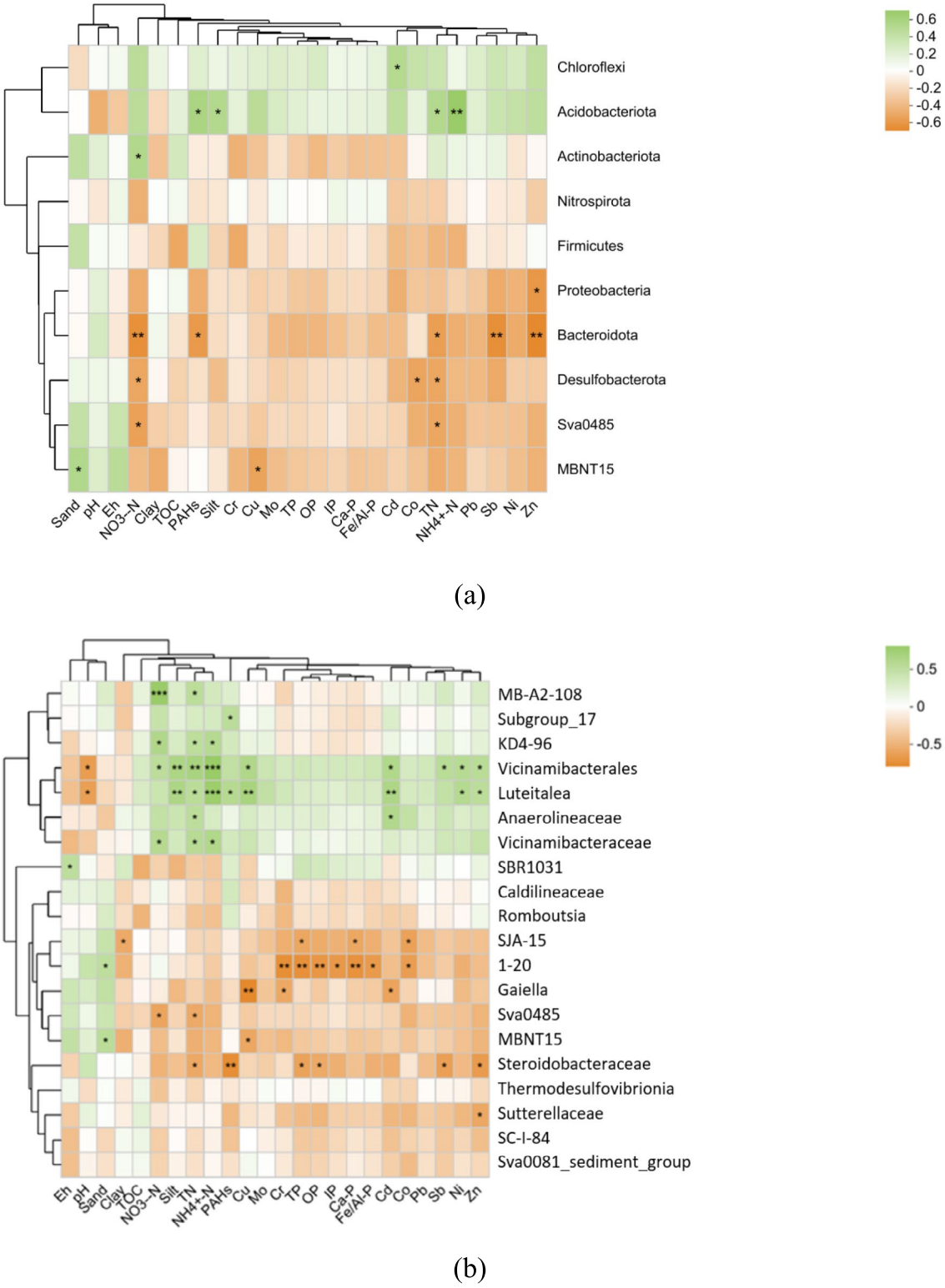

Fig. 6 Spearman correlation between environmental factors and sediment microbials on phylum level (**a**) and generic level (**b**)

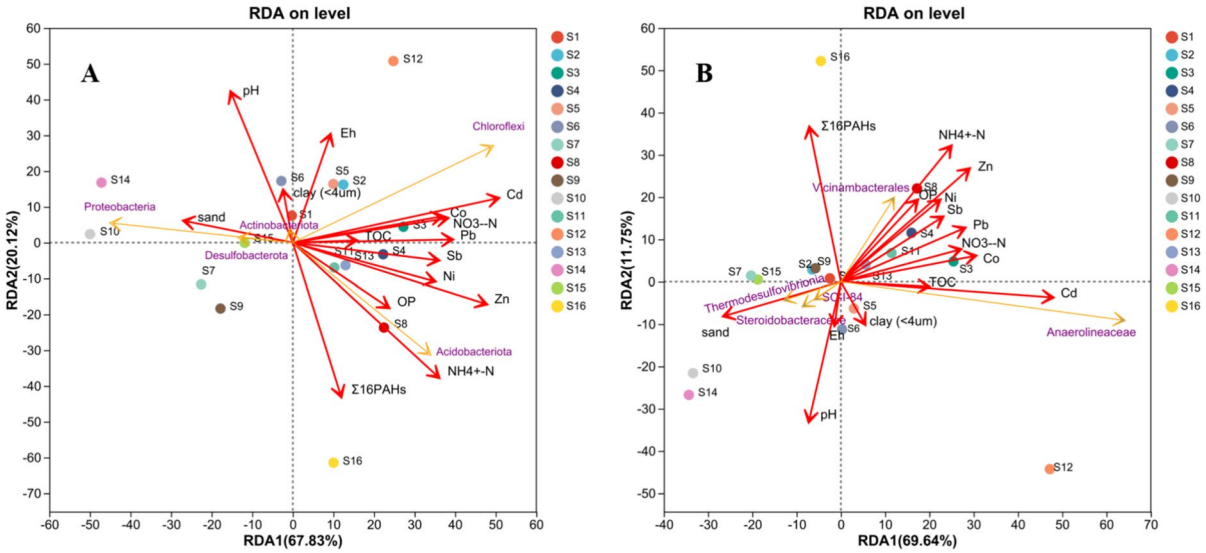

Fig. 7 Redundancy analysis (RDA) of the top five most abundant microbial (**A**) phyla and (**B**) genera in relation to influencing factors along the sediment sampling sequence in Dianshan Lake

genera showing significant negative correlations with phosphorus levels.

Multivariate relationships between environmental variables and microbial community structure

Figure 7a and b shows the RDA ordinations at the phylum and genus levels, respectively. Notably, samples S12 and S16 were the most distinct, indicating considerable differences in microbial community composition. Among the environmental factors, NH_4^+-N, PAHs, Cd, pH, Zn, and Eh contributed most substantially to variations in community structure.

At the phylum level, Acidobacteriota exhibited a positive response to NH_4^+-N, which was strongly associated with S8 but negatively correlated with S1 and S6. Cd had a pronounced influence on Chloroflexi, which also appeared to be moderately affected by Eh. Proteobacteria generally responded negatively to both PTEs and PAHs, with PAHs exerting a particularly strong effect on S16. In contrast, Desulfobacterota and Actinobacteriota showed relatively weak responses to the measured environmental factors.

At the genus level, *Anaerolineaceae* (Chloroflexi) was most strongly influenced by environmental conditions, particularly showing a positive response to Cd. *Vicinamibacterales* (Acidobacteriota) displayed a moderate positive response to NH_4^+-N but a negative response to pH.

Random Forest and SHAP analysis

A RF algorithm was applied to quantify and interpret the relative contributions of environmental variables to the Shannon diversity index. The model achieved an R^2 of 0.83 and an RMSE of 0.057, indicating robust predictive performance for sediment α-diversity. According to the factor contribution ranking (Fig. 8a), contaminants exerted the greatest influence on diversity, accounting for ~40% of the total effect, mainly contributed by Cu, Cd, and Sb. High values of Sb and Cu generally had negative contributions, whereas Cd was mostly positively associated, with the exception of a single point. Sediment physicochemical properties were the second most influential group, with Eh contributing most prominently. SHAP analysis further indicated that high-value points exerted bidirectional effects on Shannon diversity, whereas low-value points had consistently negative impacts (Figure S7). Although the total contribution of nutrient elements was the least, NH_4^+-N and NO_3^--N showed relatively high overall influence among all single factors.

Both RF feature importance and SHAP analyses were also conducted for the top five dominant

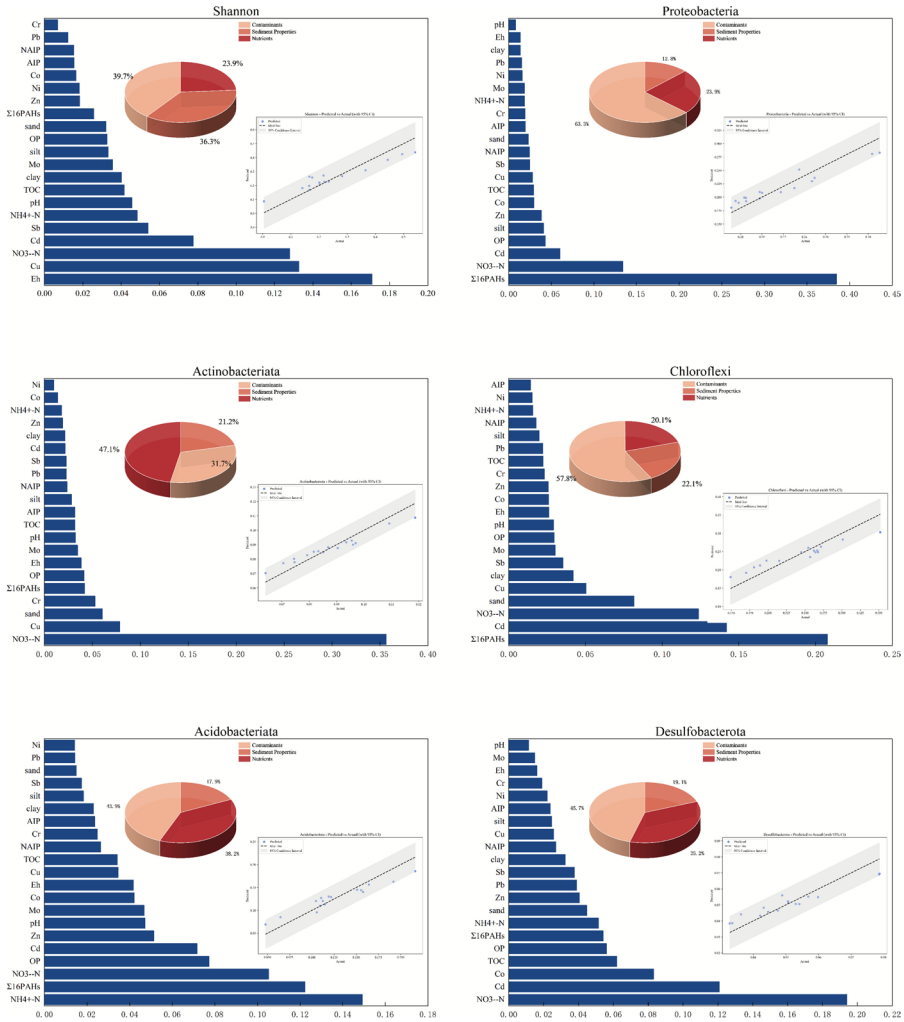

Fig. 8 Contributions of environmental factors to the Shannon diversity index (**a**) and the abundances of Proteobacteria (**b**), Actinobacteriota (**c**), Chloroflexi (**d**), Acidobacteriota (**e**), and Desulfobacterota (**f**) in surface sediments from Dianshan Lake, as estimated by the Random Forest model. Insets show the model fit, comparing modelled microbial abundances to observed values.

bacterial phyla—Proteobacteria, Actinobacteriota, Chloroflexi, Acidobacteriota, and Desulfobacterota—with results presented in Figs. 8b–f and Figure S7. The model demonstrated strong predictive performance for these phyla, with correlations ranging from 0.80 to 0.88.

Significant differences in factor contributions were observed among the phyla. Contaminants had the greatest influence on Proteobacteria and Chloroflexi, contributing over 50% of the total variation. Among contaminants, Cd and PAHs were the most impactful across all phyla, although SHAP analysis indicated that their effects varied in direction among different phyla. Nutrient factors ranked second in importance, with NO_3^--N exerting significant effects on all phyla except Acidobacteriota, where NH_4^+-N was the most influential. For Actinobacteriota, NO_3^--N accounted for over 35% of the contribution, with low-value points showing negative effects and high-value points showing bidirectional effects (Figure S7). Desulfobacterota was also strongly influenced by NO_3^--N, but in the opposite direction.

In contrast, sediment physicochemical properties contributed the least to the five dominant phyla, with variable effects across taxa. For example, Eh ranked fourth in its influence on Acidobacteriota but had minimal effect on Proteobacteria.

Discussion

The dominant phyla in the surface sediments of Dianshan Lake (Fig. 4) are largely consistent with those found in other major freshwater lakes both

domestically and internationally (Zhao et al., 2019; Chao et al., 2023; Yang et al., 2020; Chao et al., 2021; Paul et al., 2016; Borsetto et al., 2021), suggesting a broadly conserved microbial community composition. However, the spatial heterogeneity of diversity metrics indicates distinct environmental gradients and anthropogenic influences across different lake zones, contributing to the observed variability in microbial community structure. These results highlight the complex interactions between microbial communities and environmental gradients in lake sediments and underscore the sensitivity of microbial populations to nutrient and pollutant dynamics.

Effect of contaminants on microbial community

Random Forest and SHAP analyses revealed that polycyclic aromatic hydrocarbons (PAHs) had a limited overall impact on microbial diversity, likely due to their relatively low concentrations in Dianshan Lake sediments (Fig. 2). Nevertheless, PAHs did affect the abundance of specific phyla and genera. For instance, PAHs contributed most significantly to Proteobacteria (Fig. 8b), particularly negatively affecting the genus *Steroidobacteraceae* (Fig. 6b), which is the most abundant genus within this phylum. Interestingly, some taxa exhibited positive responses to PAHs, including members of Chloroflexi and Acidobacteriota (Figs. 6 and S7). Chloroflexi are capable of utilizing light energy for autotrophy and exhibit high adaptability to diverse environmental conditions. Many Chloroflexi species can also degrade organic pollutants (Zhao et al., 2019), providing them a competitive advantage in anthropogenically influenced environments. Acidobacteriota have been reported to utilize PAHs as carbon sources, with degradation potential increasing with PAH concentration (Su et al., 2017). In Dianshan Lake sediments, the primary effects of PAHs were observed in the genera *subgroup_17* and *Luteitalea* in Acidobacteriota (Fig. 6). However, these genera accounted for only 1.78% and 1.90% of the total microbial abundance, respectively (Fig. 4). Consequently, PAHs had a less pronounced effect on the overall Acidobacteriota phylum compared to Proteobacteria in the RF analysis (Fig. 8).

Potentially toxic elements (PTEs) such as Cu, Cd, and Sb significantly influenced the Shannon diversity index in Dianshan Lake sediments (Fig. 8a), with Cu and Sb exhibiting inhibitory effects on microbial diversity. Regarding community structure, PTEs affected the abundance of most dominant phyla, with Cd showing the strongest influence on Proteobacteria, Chloroflexi, Acidobacteriota, and Desulfobacterota, ranking among the top three contributors of all factors. Sb, Cu, Zn, and Co were all significantly negatively correlated with the abundance of dominant microbial phyla (Fig. 6a), indicating a generally suppressive effect of these metals. Similar trends were reported by Guo et al. (2019), where Cu and Zn discharged into freshwater lake sediments led to significant declines in Proteobacteria and Bacteroidota abundance. The toxic effects of PTEs on microorganisms are typically attributed to interactions between metal ions and functional ligands (e.g., thiol groups in enzymes), which inhibit essential enzymatic activities (Olaniran et al., 2013).

Conversely, several genera exhibited positive correlations with PTE concentrations, including *Anaerolineaceae* (Chloroflexi), *Vicinamibacterales*, and *Luteitalea* (Acidobacteriota), all of which were positively associated with Cd (Fig. 6b). Comparable results were reported by Wang et al. (2023a) in sediments downstream of a mining area in the Zijiang River in China. These responses may reflect the ability of microorganisms to tolerate PTEs below critical threshold concentrations (Giller et al., 2009). For example, Lee et al. (2024) observed that Luteitalea could withstand soils contaminated with Cu and Pb for up to 80days in pot experiments.

Effect of nutrients on microbial community

The concentrations of essential nutrients, such as carbon (C), nitrogen (N), and phosphorus (P), in sediments play a crucial role in shaping microbial diversity and community structure (Yao et al., 2022). In Dianshan Lake, C and N were the most influential factors affecting microbial diversity, whereas the effect of P was comparatively minor.

Correlation, Random Forest (RF), and SHAP analyses consistently indicated that nitrogen, particularly NO_3^--N, was a primary driver of microbial diversity and community composition in Dianshan Lake sediments (Figs. 6, 7, and S?). This is consistent with findings from several lakes in Nanjing, China (Zhao et al., 2012), and suggests that NO_3^--N supports microbial growth by promoting nitrogen

cycling and energy metabolism (Wang et al., 2022a). Nitrogen cycling appears to play a more prominent role than carbon or sulfur cycling in freshwater sediments (Virdis et al., 2011; Yoo et al., 1999; Zhang et al., 2024a; Zheng et al., 2015). At the phylum level, NO_3^--N had a strong effect on all dominant phyla except Acidobacteriota, with Actinobacteriota showing a marked positive response. Supporting evidence comes from studies in aerobic sludge reactors, where NO_3^--N addition increased Actinobacteriota abundance, which are known to harbor denitrification genes (Chen et al., 2025; Zhang et al., 2024b). Denitrification-related enzymes have also been reported in Chloroflexi, particularly within Anaerolineaceae (Feng et al., 2023; Qian et al., 2023; Wu et al., 2024a). However, the effects of NO_3^--N on microbial abundance were not uniformly positive. Similar patterns have been reported in other systems: Wang et al. (2022a) observed a negative correlation between Sva0485 abundance and NO_3^--N in estuarine sediments, and Shao et al. (2023) and Lin et al. (2022) reported opposing responses of Acidobacteriota and Bacteroidetes to pollution gradients. These discrepancies highlight the complexity of microbial community dynamics and the challenges of interpreting single-factor correlations.

For Acidobacteriota, NH_4^+-N appears to be a key driver of relative abundance, consistent with observations from coastal wetland sediments (Lin & Lin, 2022). Organic carbon is a critical energy source for chemotrophic microorganisms; consequently, the lower TOC content in northern Dianshan Lake sediments likely contributes to reduced microbial diversity in this region. TOC was found to positively influence the dominant phylum Proteobacteria, while having minimal effect on other phyla, such as Chloroflexi and Acidobacteriota, which are less sensitive to organic carbon content (Qi et al., 2024).

Effects of sediment properties on microbial community

Physicochemical properties of sediments also strongly influence microbial diversity and community structure. Redox potential (Eh) is a key integrative parameter reflecting nutrient status, pollutant levels, and biological activity (Husson, 2013). It is closely associated with microbial functional gene composition (Zhang et al., 2015b) and influences denitrification rates and overall bacterial respiration (Hunting & van der Geest, 2011). In this study, RF analysis indicated that Eh contributed most to the Shannon diversity index (Fig. 8a). Lam et al. (2018) similarly reported higher sediment biodiversity under more negative Eh values, providing greater energy availability for microbial metabolism. At the genus level, Eh positively correlated with SBR1031, a heterotrophic microorganism, suggesting that dissolved oxygen in sediment porewater promotes its growth (Hu et al., 2023).

Sediment particle size affects microbial distribution indirectly by influencing nutrient availability and facilitating gas and ion exchange in pore spaces, rather than exerting a direct effect (Lin et al., 2023). Dianshan Lake sediments are predominantly silt, with minimal particle size variation. Nevertheless, sand content showed differential effects on phyla: higher sand proportions positively influenced Actinobacteriota, likely due to higher porosity favoring aerobic and facultative aerobes, whereas anaerobic Chloroflexi were negatively affected.

Sediment pH mediates microbial composition by affecting enzyme activity, growth, and metabolism (Rousk et al., 2010; Shang et al., 2020). In Dianshan Lake, sediment pH ranged from neutral to weakly alkaline, showing no significant effect on dominant phyla. However, a significant negative correlation was observed for the genera Vicinamibacterales and Luteitalea, suggesting their preference for neutral pH conditions (Sait et al., 2006).

Combined effects on microbial communities

Microbial tolerance to PTEs is influenced by the combined effects of multiple environmental factors and interspecific interactions, making correlations between pollutants and microbial abundance complex. For example, the abundance of *Gaiella* (Actinobacteriota) shows variable responses to PTEs. Gao et al. (2021) and Wu et al. (2024b) reported higher *Gaiella* abundance in heavily polluted soils, whereas Guo et al. (2023) found suppression under long-term PTE exposure, consistent with our results.

Interactions with other microbial taxa also modulate responses: *Gaiella* may be affected by competition with dominant genera *Anaerolineaceae*, *Vicinamibacterales*, and *Luteitalea*, which exhibit higher Cd tolerance (Fan et al., 2020). *Anaerolineaceae* may

release Cd during organic matter degradation, while methane-metabolizing microbes may facilitate Cd precipitation (Meng et al., 2019). In Dianshan Lake, the spatial coincidence of TOC and PTE pollution (e.g., higher in inflow areas like Qiandengpu) provides energy to *Anaerolineaceae*, enhancing its adaptation to Cd-rich environments (Wang et al., 2023b; Zárate et al., 2021). *Vicinamibacterales* may tolerate PTEs through symbiotic interactions with other microbes and can survive in highly polluted habitats such as mining areas (Chun et al., 2021).

Conclusions

This study analyzes the environmental factors in Dianshan Lake and reveals that its sediments possess the typical physicochemical characteristics of alluvial plain lakes, with their distribution significantly influenced by inflow and outflow rivers. Substantial variability in N, P, and selected pollutants (Cd, Cu, Mo, Sb, Zn, and PAHs) was observed across the sampling sites, mainly reflecting the influence of inflow rivers and surrounding land use. The average Shannon Index of microorganisms in Dianshan Lake sediments was 6.251. The dominant species in the sediments, mainly Chloroflexi and Proteobacteria, are consistent with those found in other lakes. The co-occurrence network's relatively high average degree indicates a large and complex microbial population structure at the genus level, with *Anaerolineaceae* being the most abundant (ranging from 3.60% to 24.77%).

Correlation, RF, and SHAP analysis identified significant relationships between key environmental factors and microorganisms, showing a clear regional pattern. Redox conditions emerged as the most significant single factor influencing microbial diversity, as they determine the types of microorganisms participating in different respiratory pathways. However, the impact of environmental factors on dominant species varied significantly. PAHs had the greatest impact on Proteobacteria and Chloroflexi, mainly inhibiting the genus *Steroidobacteraceae*. RF analysis also showed that Cd can significantly affect Chloroflexi, Proteobacteria, and Desulfobacterota. Furthermore, genera such as *Anaerolineaceae* (Chloroflexi), *Vicinamibacterales* (Acidobacteriota), and *Luteitalea* (Acidobacteriota) exhibited a significant positive correlation with specific PTEs, including Cd, indicating their tolerance to contamination. The dominant species in Dianshan Lake sediments were also strongly influenced by NO_3^--N and NH_4^+-N, far exceeding the impact of various forms of phosphorus. These interactions reflect the relationships among microorganisms in Dianshan Lake sediments, particularly influenced by amensalism and competition.

In conclusion, this study demonstrates that combining the Random Forest (RF) algorithm with SHAP analysis effectively quantifies the influence of environmental factors on sediment microbial diversity and community structure. The identified key drivers in lightly to moderately polluted sediments provide a scientific basis for ecological risk management in similar aquatic systems. Nevertheless, as the results are based on single-time sampling in a low-altitude plain lake (Dianshan Lake, autumn season), future multi-seasonal studies are needed to better capture the spatiotemporal dynamics of sediment microbial communities.

Author contributions Zhiyi Yang: Conceptualization, Writing-Original Draft, Formal Analysis; Yinyan Ruan: Methodology, Software; Bokun Zhang: Methodology, Investigation, Validation; Xinyang Huang: Investigation, Data Curation; Feipeng Li: Resources, Project Administration; Lingchen Mao: Supervision, Writing-Review & Editing, Resources, Data Curation

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Data availability The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information files. Should any raw data files be needed in another format, they are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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